

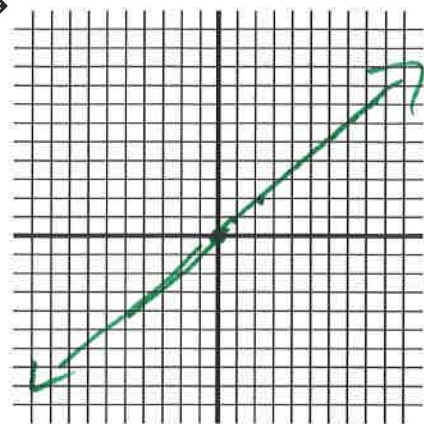
ALGEBRA 2 6.7 - INVERSE FUNCTIONS

- **Inverse functions:** Functions where the x s and y s are switched
- **Notation:** The inverse of $f(x)$ is noted $f^{-1}(x)$. The inverse of $g(x)$ is noted $g^{-1}(x)$ etc.
- **Graphs of inverses** are reflections over the line $y = x$ $\rightarrow$

$$y = mx + b$$

I. Writing Equations Of Inverse Functions. Steps:

1. Replace $f(x)$ with y
2. Replace all x s with y , and y with x
3. Solve for y
4. Replace y with $f^{-1}(x)$



Examples: Find the equation of the inverse of each function.

1. $f(x) = 2x + 5$

$$\begin{aligned} x &= 2y + 5 \\ -5 & \quad -5 \\ \hline x - 5 &= 2y \\ \frac{x - 5}{2} &= y \\ \boxed{f^{-1}(x) &= \frac{x - 5}{2}} \end{aligned}$$

2. $f(x) = 2(x + 7)^{\frac{1}{5}}$

$$\begin{aligned} x &= 2(y + 7)^{\frac{1}{5}} \\ \frac{x}{2} &= \sqrt[5]{y + 7} \\ \left(\frac{x}{2}\right)^5 &= \left(\sqrt[5]{y + 7}\right)^5 \\ \left(\frac{x}{2}\right)^5 &= y + 7 \\ -7 & \quad -7 \\ \hline \left(\frac{x}{2}\right)^5 - 7 &= y \\ \boxed{f^{-1}(x) &= \left(\frac{x}{2}\right)^5 - 7} \end{aligned}$$

3. $f(x) = \sqrt[3]{5x - 7}$

$$\begin{aligned} x^3 &= (\sqrt[3]{5y - 7})^3 \\ x^3 &= 5y - 7 \\ +7 & \quad +7 \\ \hline x^3 + 7 &= 5y \\ \frac{x^3 + 7}{5} &= y \\ \boxed{f^{-1}(x) &= \frac{x^3 + 7}{5}} \\ &= \frac{x^3}{5} + \frac{7}{5} \end{aligned}$$

4. $f(x) = \frac{(x-3)^{11}}{5}$

$$\begin{aligned} 5x &= \frac{(y-3)^{11}}{5} \\ \sqrt[11]{5x} &= \sqrt[11]{\frac{(y-3)^{11}}{5}} \\ \sqrt[11]{5x} &= y - 3 \\ +3 & \quad +3 \\ \boxed{\sqrt[11]{5x} + 3 &= y = f^{-1}(x)} \end{aligned}$$

$$5. f(x) = \frac{3x + 10}{4 - 5x}$$

$$x = \frac{3y + 10}{4 - 5y}$$

$$x(4 - 5y) = 3y + 10$$

$$4x - 5yx = 3y + 10$$

$$4x - 5yx = 3y + 10$$

$$+ 5yx \quad + 5yx$$

$$4x = 3y + 10 + 5yx$$

$$-10 \quad -10$$

$$4x - 10 = 3y + 5yx$$

factor out y:

$$4x - 10 = y(3 + 5x)$$

$$\frac{4x - 10}{3 + 5x} = \frac{y(3 + 5x)}{3 + 5x}$$

$$f^{-1}(x) = \frac{4x - 10}{3 + 5x}$$

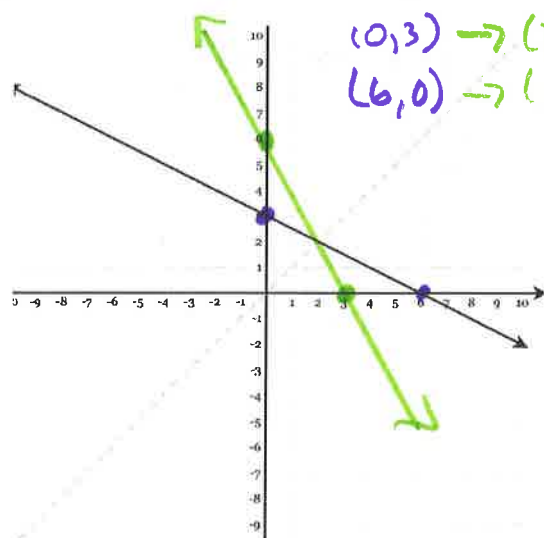
II. Finding Graphs of Inverse Functions:

- Find 2 to 5 points (depends on the graph) of the original function.
- Switch x & y for each point.
- Plot the points and sketch graph

Examples: Given the points on $f(x)$ and its graph, find points on $f^{-1}(x)$ and draw its graph.

Graph the inverse of the provided graph on the same axes.

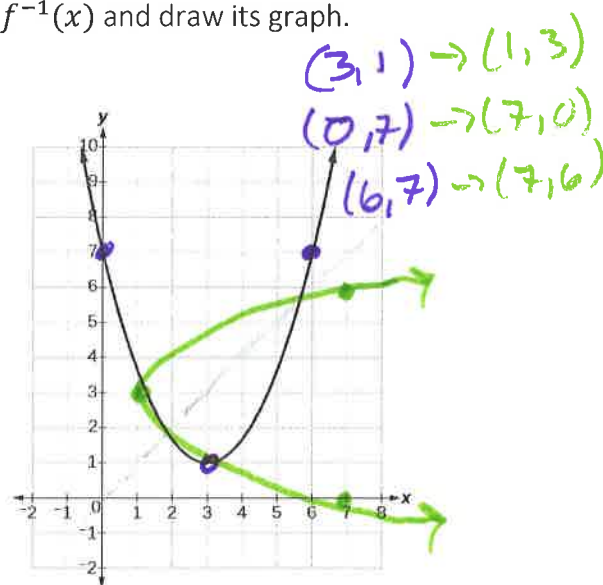
6.



$$(0, 3) \rightarrow (3, 0)$$

$$(6, 0) \rightarrow (0, 6)$$

7.



$$(3, 1) \rightarrow (1, 3)$$

$$(0, 7) \rightarrow (7, 0)$$

$$(6, 7) \rightarrow (7, 6)$$

III. The composition of a function and it's inverse: (Last thing.....SUPER easy!)

$$f(f^{-1}(x)) = x \quad \text{AND} \quad f^{-1}(f(x)) = x$$

NOTICE: A function and its inverse "undo each other"

So if written like this they cancel each other out

Examples: Evaluate.

$$8. f(f^{-1}(-5)) = \underline{-5}$$

$$9. g^{-1}(g(28)) = \underline{28}$$

$$10. g(g^{-1}(\pi)) = \underline{\pi}$$

$$11. f^{-1}(f(x)) = \underline{x}$$